

SIMATS ENGINEERING
SAVEETHA SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

ASSESSMENT TOOL – 2

SCENARIO BASED ASSESMENT

Course Code	DSA0112	Course Title	Object Oriented Programming using C++
CO Assessed	CO2 (BL3)	Assessment	Tool – 2
Weightage	20%	Total Marks	100
Student Name	Shanthosh. K	Reg No	192521251
Section / Batch	IT-2025	Date	04.08.2026
CO Statement	Design C++ programs by applying functions and object-oriented concepts such as classes, objects, access specifiers, member functions, static members, and arrays of objects to solve real-world programming problems. (BL3)		

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Code of Conduct:

*I **Shanthosh. K/192521251** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure

SCENARIO BASED ASSESSMENT

1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management

Task:

- A. Use function overloading to calculate charge for general consultation and specialist consultation.
- b. Use call by reference to modify the final bill amount.
- c. Use default arguments to add a standard service charge if needed.
- d. Use function prototyping for all calculation functions.
- e. Display the patient type and total consultation charge.(BL3)

2. Design a C++ program Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

- A. Create a class Member with attributes: Member ID, Name, Plan Type, and Fee.
- B. Implement member functions to assign plans, calculate fees, and display membership details.
- C. Use an array of objects to manage multiple members.
- D. Add validation for valid membership plans.
- E. Include a static member to track total active members.
- F. Evaluate how your design supports easy modification of pricing plans.

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System.

Task:

- A. Use function prototyping to declare EMI-related functions.
- b. Use default arguments for default interest rate where applicable.
- c. Apply call by reference to update payable amount after subsidy.
- d. Use inline functions for simple interest calculations.
- e. Display the loan amount, interest, and repayment details clearly.

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a **C++ program** using **functions and object-oriented programming concepts** to develop a **Parcel Charge Management System**.

- a. Create a **class Parcel** with appropriate **private data members** such as parcel type, weight, distance, and packaging charge.
- b. Use **public member functions** to read parcel details, calculate the delivery charge, and display the parcel information.
- c. Use **function overloading** to calculate charges for **Normal Delivery** and **Express Delivery**.
- d. Use **default arguments** to assign a default packaging charge when it is not specified by the user.
- e. Use an **inline function** to calculate the base charge based on parcel weight and delivery distance.
- f. Maintain the **total number of parcels processed** using a **static data member** and display it through a **static member function**.

5. A utility company wants to automate the calculation of water bills for **domestic** and **commercial** consumers. Design and implement a **C++ program** using **functions and object-oriented programming concepts** to develop a **Water Billing System**. Create a **class Consumer** with appropriate **private data members** such as consumer ID, consumer type, water consumption (in liters), and bill amount.
- b. Use **public member functions** to input consumer details, calculate the water bill, and display the billing information.
 - c. Implement **function overloading** to calculate water bills for **Domestic** and **Commercial** consumers using different tariff rates.
 - d. Use **call by reference** to update the bill amount after applying a government subsidy.
 - e. Use **default arguments** to include a service tax, where a default tax rate is applied unless specified.
 - f. Declare all functions using **function prototyping** before their definitions.
 - g. Maintain the **total number of consumers processed** using a **static data member** and display it through a **static member function**.
 - h. Create an **array of objects** to store and process the details of multiple consumers, and display the consumer type, water consumption, subsidy-adjusted bill amount.

Answers:

Q1. Design a C++ program for a **Hospital Consultation Charge System** using **function overloading**, **call by reference**, **default arguments**, and **function prototyping** to calculate consultation charges for general and specialist patients, and display the patient type and total consultation charge.

```
[*] 1.cpp
1  #include <iostream>
2  using namespace std;
3
4  // Function Prototypes
5  double calculateCharge(double generalFee, double serviceCharge = 100);
6  double calculateCharge(double specialistFee, double specialistCharge, double serviceCharge);
7  void modifyBill(double &bill);
8
9  int main()
10 {
11     int choice;
12     double bill;
13
14     cin >> choice;
15
16     if (choice == 1)
17     {
18         double fee;
19         cin >> fee;
20
21         bill = calculateCharge(fee);
22         modifyBill(bill);
23
24         cout << "Patient Type: General" << endl;
25         cout << "Total Consultation Charge: " << bill;
26     }
27     else if (choice == 2)
28     {
29         double fee, specialistCharge;
30         cin >> fee >> specialistCharge;
31
32         bill = calculateCharge(fee, specialistCharge, 100);
33         modifyBill(bill);
34
35         cout << "Patient Type: Specialist" << endl;
36         cout << "Total Consultation Charge: " << bill;
37     }
38     else
39     {
40         cout << "Invalid Choice";
41     }
42
43     return 0;
44 }
45
46 double calculateCharge(double generalFee, double serviceCharge)
47 {
48     return generalFee + serviceCharge;
49 }
50
51 double calculateCharge(double specialistFee, double specialistCharge, double serviceCharge)
52 {
53     return specialistFee + specialistCharge + serviceCharge;
54 }
55
56 void modifyBill(double &bill)
57 {
58     bill = bill + (bill * 0.05);
59 }
```

Output:

```
===== Hospital Consultation Charge System =====
1. General Consultation
2. Specialist Consultation
Enter your choice: 1
Enter General Consultation Fee: 500

Patient Type : General
Total Consultation Charge : Rs. 630

-----
Process exited after 20.14 seconds with return value 0
Press any key to continue . . . |
```

Q2. Design a C++ program for a Fitness Centre Membership System using a class, array of objects, static member, and member functions to assign plans, calculate membership fees, validate membership plans, and display member details.

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Member
6  {
7  private:
8      int memberID;
9      string name;
10     string planType;
11     double fee;
12
13     static int totalMembers;
14
15 public:
16     void assignPlan()
17     {
18         cout << "Enter Member ID: ";
19         cin >> memberID;
20
21         cout << "Enter Member Name: ";
22         cin >> name;
23
24         cout << "Enter Plan (Basic/Premium/VIP): ";
25         cin >> planType;
26
27         if (planType != "Basic" && planType != "Premium" && planType != "VIP")
28         {
29             cout << "Invalid Membership Plan!" << endl;
30             planType = "Basic";
31         }
32
33         calculateFee();
34         totalMembers++;
35     }
36
37     void calculateFee()
38     {
39         if (planType == "Basic")
40             fee = 1000;
41         else if (planType == "Premium")
42             fee = 2000;
43         else if (planType == "VIP")
44             fee = 3000;
45     }
```

```

46
47 void display()
48 {
49     cout << "\nMember ID : " << memberID << endl;
50     cout << "Name       : " << name << endl;
51     cout << "Plan Type : " << planType << endl;
52     cout << "Fee       : Rs. " << fee << endl;
53 }
54
55 static void displayTotalMembers()
56 {
57     cout << "\nTotal Active Members: " << totalMembers << endl;
58 }
59 };
60
61 int Member::totalMembers = 0;
62
63 int main()
64 {
65     int n;
66
67     cout << "Enter Number of Members: ";
68     cin >> n;
69
70     Member members[100];
71
72     for (int i = 0; i < n; i++)
73     {
74         cout << "\nEnter Details for Member " << i + 1 << endl;
75         members[i].assignPlan();
76     }
77
78     cout << "\n----- Membership Details -----" << endl;
79
80     for (int i = 0; i < n; i++)
81     {
82         members[i].display();
83     }
84
85     Member::displayTotalMembers();
86
87     return 0;
88 }

```

Output:

```

Enter Number of Members: 1

Enter Details for Member 1
Enter Member ID: M001
Enter Member Name: Enter Plan (Basic/Premium/VIP): Invalid Membership Plan!

----- Membership Details -----

Member ID : 0
Name       :
Plan Type : Basic
Fee        : Rs. 1000

Total Active Members: 1

-----
Process exited after 9.616 seconds with return value 0
Press any key to continue . . . |

```

Q3. Design a C++ program for a Loan Repayment System using function prototyping, default arguments, call by reference, and inline functions to calculate loan repayment details and display the loan amount, interest, and repayment amount.

```
1  #include <iostream>
2  using namespace std;
3
4  // Function Prototypes
5  double calculateEMI(double loanAmount, int years, double rate = 8.5);
6  void applySubsidy(double &payableAmount);
7  inline double calculateInterest(double loanAmount, double rate, int years);
8
9  int main()
10 {
11     double loanAmount, interest, totalAmount;
12     int years;
13
14     cout << "Enter Loan Amount: ";
15     cin >> loanAmount;
16
17     cout << "Enter Loan Period (Years): ";
18     cin >> years;
19
20     interest = calculateInterest(loanAmount, 8.5, years);
21     totalAmount = loanAmount + interest;
22
23     applySubsidy(totalAmount);
24
25     cout << "\nLoan Amount : Rs. " << loanAmount << endl;
26     cout << "Interest      : Rs. " << interest << endl;
27     cout << "Repayment Amount : Rs. " << totalAmount << endl;
28
29     return 0;
30 }
31
32 double calculateEMI(double loanAmount, int years, double rate)
33 {
34     double interest = (loanAmount * rate * years) / 100;
35     return (loanAmount + interest) / (years * 12);
36 }
37
38 inline double calculateInterest(double loanAmount, double rate, int years)
39 {
40     return (loanAmount * rate * years) / 100;
41 }
42
43 void applySubsidy(double &payableAmount)
44 {
45     payableAmount = payableAmount - 5000;
46 }
```

Output:

```
Enter Loan Amount: 200000
Enter Loan Period (Years): 2

Loan Amount : Rs. 200000
Interest      : Rs. 34000
Repayment Amount : Rs. 229000

-----
Process exited after 11.6 seconds with return value 0
Press any key to continue . . . |
```

Q4. Design a C++ program for a Parcel Charge Management System using a class, function overloading, default arguments, inline functions, static data members, and member functions to calculate and display parcel delivery charges for Normal and Express delivery.

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Parcel
6  {
7  private:
8      string parcelType;
9      float weight;
10     float distance;
11     float packagingCharge;
12
13     static int totalParcels;
14
15 public:
16     void getDetails(float packCharge = 50)
17     {
18         cout << "Enter Parcel Type (Normal/Express): ";
19         cin >> parcelType;
20
21         cout << "Enter Weight (kg): ";
22         cin >> weight;
23
24         cout << "Enter Distance (km): ";
25         cin >> distance;
26
27         packagingCharge = packCharge;
28         totalParcels++;
29     }
30
31     inline float baseCharge()
32     {
33         return weight * distance * 2;
34     }
35
36     float calculateCharge()
37     {
38         return baseCharge() + packagingCharge;
39     }
40
41     float calculateCharge(float expressCharge)
42     {
43         return baseCharge() + packagingCharge + expressCharge;
44     }
45
46     void display()
47     {
48         float total;
```



```

49
50     if (parcelType == "Express")
51         total = calculateCharge(100);
52     else
53         total = calculateCharge();
54
55     cout << "\nParcel Type      : " << parcelType << endl;
56     cout << "Weight          : " << weight << " kg" << endl;
57     cout << "Distance       : " << distance << " km" << endl;
58     cout << "Packaging Charge : Rs. " << packagingCharge << endl;
59     cout << "Total Charge    : Rs. " << total << endl;
60 }
61
62 static void displayTotalParcels()
63 {
64     cout << "\nTotal Parcels Processed: " << totalParcels << endl;
65 }
66 };
67
68 int Parcel::totalParcels = 0;
69
70 int main()
71 {
72     int n;
73
74     cout << "Enter Number of Parcels: ";
75     cin >> n;
76
77     Parcel p[100];
78
79     for (int i = 0; i < n; i++)
80     {
81         cout << "\nEnter Details for Parcel " << i + 1 << endl;
82         p[i].getDetails();
83     }
84
85     cout << "\n----- Parcel Details -----" << endl;
86
87     for (int i = 0; i < n; i++)
88     {
89         p[i].display();
90     }
91
92     Parcel::displayTotalParcels();
93
94     return 0;
95 }

```

Output:

```

Enter Number of Parcels: 1

Enter Details for Parcel 1
Enter Parcel Type (Normal/Express): Normal
Enter Weight (kg): 10
Enter Distance (km): 2000

----- Parcel Details -----

Parcel Type      : Normal
Weight          : 10 kg
Distance       : 2000 km
Packaging Charge : Rs. 50
Total Charge    : Rs. 40050

Total Parcels Processed: 1

-----
Process exited after 21.29 seconds with return value 0
Press any key to continue . . . |

```

Q5. Design a C++ program for a Water Billing System using a class, function overloading, call by reference, default arguments, function prototyping, static data members, and an array of objects to calculate and display water bills for domestic and commercial consumers.

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class Consumer
6  {
7  private:
8      int consumerID;
9      string consumerType;
10     double waterConsumption;
11     double billAmount;
12
13     static int totalConsumers;
14
15 public:
16     void inputDetails();
17     void calculateBill();
18     void calculateBill(double commercialRate);
19     void display();
20     static void displayTotalConsumers();
21 };
22
23 int totalTax = 50;
24
25 // Function Prototypes
26 void applySubsidy(double &bill);
27 double addServiceTax(double bill, double tax = 50);
28
29 int Consumer::totalConsumers = 0;
30
31 void Consumer::inputDetails()
32 {
33     cout << "Enter Consumer ID: ";
34     cin >> consumerID;
35
36     cout << "Enter Consumer Type (Domestic/Commercial): ";
37     cin >> consumerType;
38
39     cout << "Enter Water Consumption (Liters): ";
40     cin >> waterConsumption;
41
42     if (consumerType == "Domestic")
```

```

43         calculateBill();
44     else
45         calculateBill(0.05);
46
47     totalConsumers++;
48 }
49
50 void Consumer::calculateBill()
51 {
52     billAmount = waterConsumption * 0.02;
53     billAmount = addServiceTax(billAmount);
54     applySubsidy(billAmount);
55 }
56
57 void Consumer::calculateBill(double commercialRate)
58 {
59     billAmount = waterConsumption * commercialRate;
60     billAmount = addServiceTax(billAmount);
61     applySubsidy(billAmount);
62 }
63
64 void Consumer::display()
65 {
66     cout << "\nConsumer ID : " << consumerID << endl;
67     cout << "Consumer Type : " << consumerType << endl;
68     cout << "Water Consumption : " << waterConsumption << " Liters" << endl;
69     cout << "Bill Amount : Rs. " << billAmount << endl;
70 }
71
72 void Consumer::displayTotalConsumers()
73 {
74     cout << "\nTotal Consumers Processed: " << totalConsumers << endl;
75 }
76
77 void applySubsidy(double &bill)
78 {
79     bill = bill - 100;
80     if (bill < 0)
81         bill = 0;
82 }
83
84 double addServiceTax(double bill, double tax)

```

```

85 {
86     return bill + tax;
87 }
88
89 int main()
90 {
91     int n;
92
93     cout << "Enter Number of Consumers: ";
94     cin >> n;
95
96     Consumer c[100];
97
98     for (int i = 0; i < n; i++)
99     {
100         cout << "\nEnter Details for Consumer " << i + 1 << endl;
101         c[i].inputDetails();
102     }
103
104     cout << "\n----- Consumer Details -----" << endl;
105
106     for (int i = 0; i < n; i++)
107     {
108         c[i].display();
109     }
110
111     Consumer::displayTotalConsumers();
112
113     return 0;
114 }

```

Outputs:

```

Enter Number of Consumers: 2

Enter Details for Consumer 1
Enter Consumer ID: 101
Enter Consumer Type (Domestic/Commercial): Commercial
Enter Water Consumption (Liters): 7500

Enter Details for Consumer 2
Enter Consumer ID: 102
Enter Consumer Type (Domestic/Commercial): Domestic
Enter Water Consumption (Liters): 5700

----- Consumer Details -----

Consumer ID : 101
Consumer Type : Commercial
Water Consumption : 7500 Liters
Bill Amount : Rs. 325

Consumer ID : 102
Consumer Type : Domestic
Water Consumption : 5700 Liters
Bill Amount : Rs. 64

Total Consumers Processed: 2

-----
Process exited after 51.96 seconds with return value 0
Press any key to continue . . . |

```